CRC cards

CRC cards for 301 project

UserProfile

1. Keep the user ID, name, email, phone, alert setting, and photo ID.

2. Remember if the user is blocked from owner actions.

3. Let the same account use entrant, owner, or admin pages when allowed.

`FirebaseRepository`

`Campsite`

`Reservation`

`Photo`

Campsite

1. Keep name, location, GPS point, guest limit, price, rules, tags, rating, and photo ID.

2. Keep which nights the owner made availble.

3. Show info to entrants, and allow its owner to edit it.

`UserProfile`

`Lottery`

`Reservation`

`BookedNight`

`Photo`

`Rating`

Lottery

1. Keep the campsite IDs, registration start/end, booking start/end, winner count, and waitlist limit.

2. Show if joining is open or closed.

3. Keep a status so the draw happens only once.

`Campsite`

`WaitlistEntry`

`BookingOffer`

`LotteryWorker`

WaitlistEntry

1. Keep one user ID and one lottery ID.

2. Keep joined, left, won, or lost status.

3. Stop the same user from joining the same lottery twice.

`UserProfile`

`Lottery`

`FirebaseRepository`

`LotteryWorker`

BookingOffer

1. Store which winning user may book and which lottery they won.

2. Remember offer status: open, used, declined, or expired.

3. Allow choosing only an offered campsite and free dates.

`Lottery`

`UserProfile`

`Reservation`

`FirebaseRepository`

Reservation

1. Keep the entrant, campsite, guest count, nights, and status.

2. Save that the entrant accepted the rules.

3. Show a confirmation number and active, past, cancelled, or rejected state.

`BookingOffer`

`Campsite`

`BookedNight`

`UserProfile`

BookedNight

1. Represent one campsite night with a unique ID, like campsite123_2027-05-15.

2. Make sure two confirmed bookings cannot take the same night.

3. Free the night when the booking is cancelled or rejected.

`Campsite`

`Reservation`

`FirebaseRepository`

AppNotification

1. Store who gets the alert and what it says.

2. Remember whether it should be pushed and if it was sent.

3. Keep records for owners and admins when allowed.

`UserProfile`

`Reservation`

`LotteryWorker`

OwnerMessage

1. Store a text message sent by an owner to a booked entrant.

2. Tie the message to a reservation.

3. Allow the recipient and admin to view it; no entrant reply button.

`UserProfile`

`Reservation`

`FirebaseRepository`

Photo

1. Keep a compressed JPG/WebP image in a Firestore document.

2. Remember whether it belongs to a profile or campsite.

3. Allow its owner or an admin to remove it.

`UserProfile`

`Campsite`

`FirebaseRepository`

Rating

1. Keep one 1–5 score per user per campsite.

2. Help display an average rating on campsite cards.

`UserProfile`

`Campsite`

`FirebaseRepository`

FirebaseRepository

1. Be the one main place where Kotlin reads/writes Firestore and gets the current Firebase user.

2. Use transactions when joining waitlists and booking nights.

3. Check owner/admin permissions in the UI and also rely on Firestore security rules.

4. Load campsite data, users, offers, photos, alerts, ratings, and messages.

`UserProfile`

`Campsite`

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LotteryWorker

1. Run from GitHub Actions around once per hour using the Firebase Admin SDK.

2. Find closed lotteries that have not been drawn.

3. Choose unique winners, save winner/loser records, and create booking offers.

4. Send pending FCM notifications and save results without exposing secret keys in the app.

`Lottery`

`WaitlistEntry`

`BookingOffer`

`AppNotification`